TikTok Virality Prediction

Audio + Lyrics Combined Model — Feature & Results Report

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1. DATASETS

Dataset	Songs	Label	Positives	Negatives	Songs w/ Lyrics
Audio TikTok	2,771	On US TikTok chart (2021–2026)	1,625 (59%)	1,146 (41%)	2,188 (79%)
Audio Billboard	4,111	Appeared on Hot 100 (2014–2024)	3,237 (79%)	874 (21%)	3,875 (94%)

2. AUDIO FEATURES (102 USED)

Source: friend's audio extraction pipeline (readable features layer). Two features excluded: af_duration_sec and af_active_frame_fraction (file-level, not musical).

Group	Features
Rhythm / Tempo (9)	af_tempo_bpm · af_beat_interval_cv · af_onset_rate_hz · af_onset_strength_mean · af_onset_strength_std · af_onset_strength_early_mean · af_onset_strength_middle_mean · af_onset_strength_late_mean · af_rms_temporal_centroid_fraction
Loudness / Energy (20)	af_rms_linear_mean · af_rms_db_mean · af_rms_linear_std · af_rms_db_std · af_rms_db_p10 · af_rms_db_p50 · af_rms_db_p95 · af_rms_db_p95_minus_p10 · af_peak_db · af_crest_factor_db · af_rms_db_early_mean · af_rms_db_middle_mean · af_rms_db_late_mean · af_rms_db_late_minus_early · af_strongest_local_rms_change_db · af_largest_local_rms_rise_db · af_largest_local_rms_drop_db · af_onset_strength_mean (energy proxy) · af_rms_db_p10 · af_rms_db_p95
Spectral (16)	af_spectral_centroid_mean_hz · af_spectral_centroid_std_hz · af_spectral_bandwidth_mean_hz · af_spectral_bandwidth_std_hz · af_spectral_rolloff_mean_hz · af_spectral_rolloff_std_hz · af_spectral_flatness_mean · af_spectral_flatness_std · af_zcr_mean · af_zcr_std · af_spectral_contrast_mean_db · af_spectral_contrast_std_db · af_spectral_entropy_mean · af_spectral_entropy_std · af_spectral_flux_mean · af_spectral_flux_std
Frequency Bands (5)	af_bass_power_fraction · af_mid_power_fraction · af_high_power_fraction · af_spectral_centroid_mean_hz · af_spectral_rolloff_mean_hz
Timbre / MFCC (26)	af_mfcc_1_mean/std through af_mfcc_13_mean/std (26 coefficients)
Chroma / Key (27)	af_chroma_0_mean/std through af_chroma_11_mean/std · af_mean_chroma_entropy · af_dominant_pitch_class_concentration · af_key_strength
Mood — experimental (2)	af_valence · af_arousal (CLAP embeddings → small regression model, ~0.87 corr with Cyanite)
Psychoacoustic (2)	af_zwicker_equal_parent_rms_mean_sone · af_sharpness_equal_parent_rms_mean_acum

3. LYRIC FEATURES (51 TOTAL)

lf_ Basic Song-Level (11)

Feature	Description
lf_vocab_richness	Unique tokens / total tokens
lf_rhyme_density	Fraction of adjacent line pairs sharing last 3 chars
lf_hook_strength	Most-repeated line count / total lines
lf_energy_score	Energy/hype word density (fire, lit, wild ...)
lf_positive_sent	Positive emotion word density

Feature	Description
lf_negative_sent	Negative/dark emotion word density
lf_repetition_rate	Fraction of lines that repeat 2x or more
lf_chorus_density	Chorus/hook sections / total sections
lf_avg_line_length	Average words per line (normalised 0-1)
lf_verse_density	Verse sections / total sections
lf_compression_ratio	Unique lines / total lines

lf2_ Section-Level Signals (17)

Feature	Description
lf2_hook_vocabulary	Hook keyword density in best section
lf2_cross_song_repetition	Lines repeated across full song (true chorus lines)
lf2_internal_repetition	Bigram/trigram repeats within best section
lf2_sentiment_extremity	Absolute VADER compound score of best section
lf2_end_rhyme_density	Fraction of end-word pairs that rhyme
lf2_line_conciseness	Short punchy lines (5-8 words peak) — TikTok clip friendly
lf2_syllable_consistency	Low CV of syllable counts per line = consistent rhythm
lf2_phonetic_openness	Vowel density — open sounds easier to sing/sustain
lf2_pronoun_relatability	I/me/you/we density — personal = shareable
lf2_internal_rhyme	Fraction of lines with non-adjacent rhyming word pairs
lf2_energy_markers	!/? and ALL-CAPS word count per line
lf2_best_section_viral_score	Weighted viral score of best-ranked section
lf2_top_line_score	Viral score of single highest-scoring lyric line
lf2_avg_line_score	Mean viral score across all lyric lines
lf2_sentiment_compound	VADER compound score of full song (-1 to +1)
lf2_n_sections	Number of distinct labeled sections
lf2_n_lyric_lines	Total lyric lines in the song

lf3_ Tier 1 NLP (12)

Feature	Description
lf3_flesch_reading_ease	Flesch readability score — higher = more accessible
lf3_avg_syllables_per_line	Average syllable count per lyric line
lf3_std_syllables_per_line	Std deviation of syllables/line — measures flow regularity
lf3_alliteration_density	Adjacent word pairs sharing first consonant
lf3_imperative_ratio	Lines starting with action verbs (go, run, dance ...)
lf3_question_density	Lines containing a question mark
lf3_explicit_density	Slang/explicit word density
lf3_lexical_density	Content words / total words ratio
lf3_line_semantic_cohesion	Avg TF-IDF cosine similarity between adjacent lines
lf3_opening_line_score	Viral score proxy of the first lyric line
lf3_closing_line_score	Viral score proxy of the last lyric line
lf3_unique_rhyme_sounds	Number of distinct end-rhyme sounds (normalised)

lf_emb_ Sentence Embeddings + Contrast (11)

Feature	Description
lf_emb_0 ... lf_emb_9	PCA(10) of all-MiniLM-L6-v2 sentence-transformer embeddings (384-dim reduced to 10 components, 31% variance explained). Captures semantic meaning of full lyrics.
lf_chorus_verse_contrast	Cosine distance between chorus and verse embeddings — higher = more contrast between sections

4. MODELS TRAINED

Model	Type	Key Hyperparameters
Logistic Regression	Linear	L2 penalty · C=1.0 · lbfgs solver · max_iter=1000 · class_weight=balanced (Billboard)
Random Forest	Ensemble (Bagging)	n_estimators=300 · max_depth=None · min_samples_leaf=5 · n_jobs=-1
Gradient Boosting	Ensemble (Boosting)	n_estimators=300 · learning_rate=0.05 · max_depth=4 · subsample=0.8

Pipeline (all models)

Median Imputation → StandardScaler → Classifier → Platt Calibration (sigmoid, prefit)

Cross-Validation

Artist-grouped 5-fold CV · 3 repeats · Artists shuffled with seed=repeat, dealt round-robin into folds · AUC reported as mean +/- std across 15 folds · 95% CI from bootstrap (1,000 resamples) on OOF scores (repeat 0).

5. TRAINING RESULTS

TikTok Model

Features (n)	Model	Mean AUC	+/- Std	95% CI
Audio only (102)	Logistic Regression	0.762	0.026	[0.737, 0.773]
Audio only (102)	Random Forest	0.768	0.036	[0.743, 0.778]
Audio only (102)	Gradient Boosting	0.777	0.027	[0.754, 0.789]
Lyrics only (51)	Logistic Regression	0.781	0.018	[0.763, 0.797]
Lyrics only (51)	Random Forest	0.808	0.014	[0.792, 0.824]
Lyrics only (51)	Gradient Boosting	0.802	0.016	[0.786, 0.820]
Combined (153)	Logistic Regression	0.827	0.017	[0.813, 0.843]
Combined (153)	Random Forest	0.830	0.018	[0.813, 0.844]
Combined (153) BEST	Gradient Boosting	0.842	0.015	[0.828, 0.857]

BEST — saved as combined_tiktok_combined.joblib
Friend's audio-only baseline (readable + tags): AUC 0.80 — combined model improvement: +4.2 points

Billboard Model

Features (n)	Model	Mean AUC	+/- Std	95% CI
Audio only (102)	Logistic Regression	0.620	0.033	[0.601, 0.643]
Audio only (102)	Random Forest	0.632	0.034	[0.603, 0.646]
Audio only (102)	Gradient Boosting	0.618	0.027	[0.591, 0.634]

Features (n)	Model	Mean AUC	+/- Std	95% CI
Lyrics only (51)	Logistic Regression	0.642	0.024	[0.627, 0.666]
Lyrics only (51)	Random Forest	0.656	0.025	[0.637, 0.676]
Lyrics only (51)	Gradient Boosting	0.642	0.026	[0.617, 0.656]
Combined (153)	Logistic Regression	0.668	0.027	[0.657, 0.694]
Combined (153) BEST	Random Forest	0.670	0.034	[0.650, 0.689]
Combined (153)	Gradient Boosting	0.659	0.025	[0.639, 0.677]

BEST — saved as combined_billboard_combined.joblib
Friend's audio-only baseline (readable only): AUC 0.61 — combined model improvement: +6.0 points

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6. KEY FINDINGS

Finding	Detail
Lyrics beat audio on TikTok	Lyrics-only AUC 0.808 > Audio-only 0.777 — lyric signals are more predictive of TikTok virality than audio features alone.
Combined always wins	Audio + Lyrics combined outperforms either modality alone on both tasks.
TikTok improved +4.2 pts vs friend's model	Friend's audio model (readable + tags): 0.80 — Combined: 0.842.
Billboard improved +6.0 pts vs baseline	Audio baseline 0.61 — Combined 0.670. Billboard is harder to predict (charted songs are more famous, harder to separate from audio/lyrics alone).
Best model per task	TikTok: Gradient Boosting (153 features). Billboard: Random Forest (153 features).
OOF scores available for stacking	Saved to results/ — can be combined with friend's model OOF scores for a stacked ensemble.